SOLUTION

Experiment 2 — Reinforcement Learning: Q-Learning

4×4 Grid World Environment

1. Objective

To implement Q-Learning in a 4×4 Grid World and observe how the agent learns an effective route to the goal while considering step and obstacle penalties.

2. Environment

The environment contains 16 states arranged as a 4×4 grid. State 0 is the starting state, state 15 is the goal, and states 5 and 10 are obstacles. The action space contains Up, Down, Left and Right.

3. Reward Structure

Reaching the goal gives +10. Every normal movement gives −1. Attempting to enter an obstacle gives −5. Boundary moves are treated as invalid moves and receive −1 while keeping the agent in the same state.

4. Hyperparameters

Learning rate $\alpha = 0.10$; discount factor $\gamma = 0.90$; exploration rate $\varepsilon = 0.20$; training episodes = 100. The Q-table is initialized with zeros.

5. Q-Learning Update

The update rule used is $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max Q(s',a') - Q(s,a)]$. For a terminal goal transition, the future-value term is set to zero.

6. Q-Table Evolution

Snapshots are recorded after Episodes 1, 50 and 100. Early values are mostly close to zero because the agent has little experience. By Episode 50, useful action values propagate backward from the goal. By Episode 100, actions leading toward the goal generally have larger values.

7. Final Policy

The learned policy is extracted using argmax over the four Q-values at each non-terminal state. In the final run, the policy grid is: R | R | D | D / D | X | R | D / R | D | X | D / R | R | R | G. G denotes the goal and X denotes an obstacle.

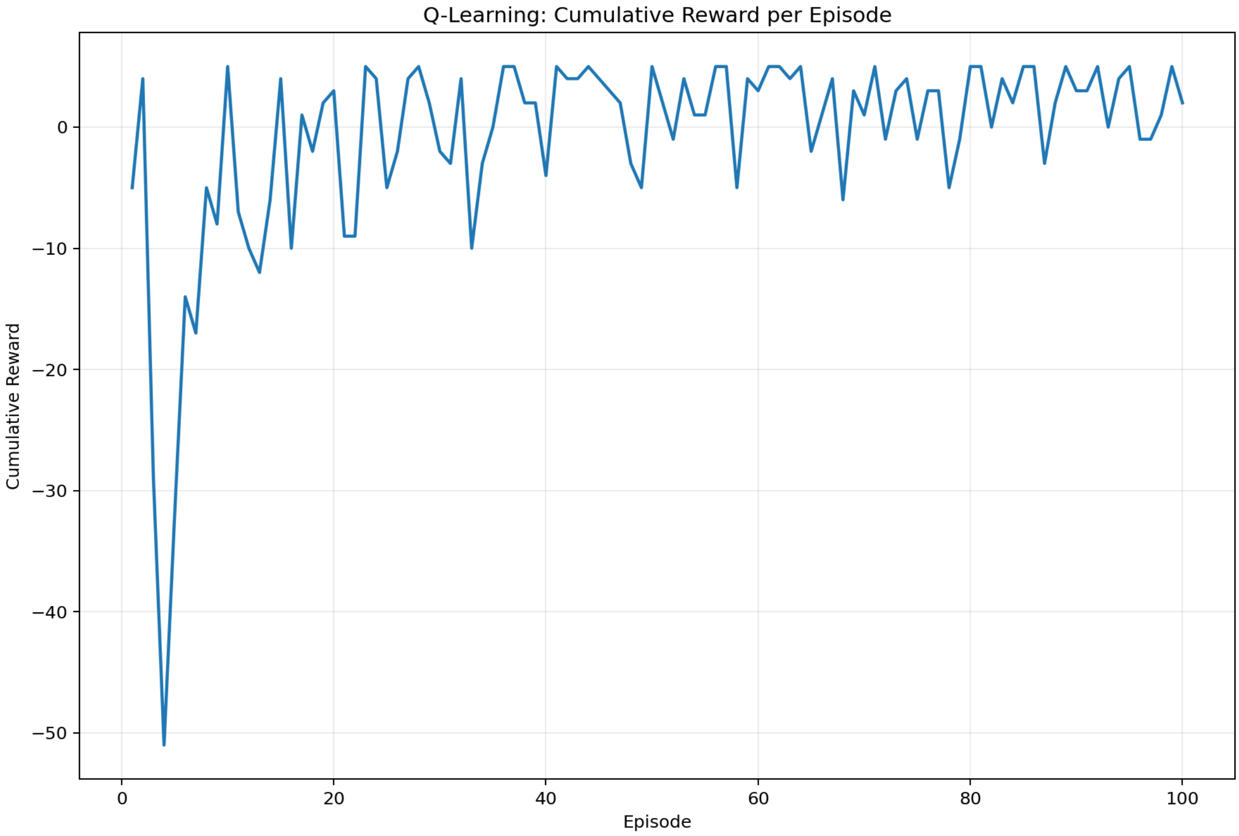
8. Convergence Behaviour

The cumulative reward is expected to improve as the agent discovers shorter and safer routes. Initial episodes may have low or negative rewards because exploration is high and the Q-table is uninformative. With repeated updates, valuable actions receive stronger Q-values and the policy becomes more consistent. Because ε remains fixed at 0.20, some exploration continues, so the reward curve may fluctuate rather than become perfectly flat.

9. Conclusion

Q-Learning successfully learns an action policy from interaction with the Grid World without requiring a pre-defined optimal policy. The experiment demonstrates exploration, exploitation, reward propagation and policy improvement.

Cumulative Reward



Q-Table Evolution